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(54) **KRAS G12C INHIBITORS AND METHODS OF USING THE SAME**

KRAS-G12C-INHIBITOREN UND VERFAHREN ZU DEREN VERWENDUNG

INHIBITEURS DE KRAS G12C ET LEURS PROCÉDÉS D'UTILISATION

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(56) References cited:
WO-A1-2015/054572 **WO-A1-2019/213516**

Description**FIELD OF INVENTION**

[0001] Provided herein are KRAS G12C inhibitors, composition of the same, and methods of using the same. These inhibitors are useful for use in the treatment of a number of disorders, including pancreatic, colorectal, and lung cancers.

Background

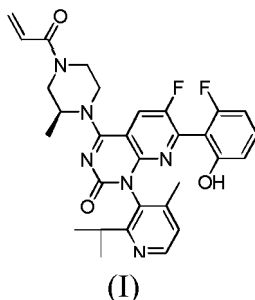
[0002] KRAS gene mutations are common in pancreatic cancer, lung adenocarcinoma, colorectal cancer, gall bladder cancer, thyroid cancer, and bile duct cancer. KRAS mutations are also observed in about 25% of patients with NSCLC, and some studies have indicated that KRAS mutations are a negative prognostic factor in patients with NSCLC. Recently, V-Ki-ras2 Kirsten rat sarcoma viral oncogene homolog (KRAS) mutations have been found to confer resistance to epidermal growth factor receptor (EGFR) targeted therapies in colorectal cancer; accordingly, the mutational status of KRAS can provide important information prior to the prescription of TKI therapy. Taken together, there is a need for new medical treatments for patients with pancreatic cancer, lung adenocarcinoma, or colorectal cancer, especially those who have been diagnosed to have such cancers characterized by a KRAS mutation, and including those who have progressed after chemotherapy.

[0003] WO2015054572 A1 discloses compounds having activity as inhibitors of G12C mutant KRAS protein.

[0004] WO2019213516A1 discloses KRAS G12C inhibitors, composition of the same, and methods of using the same.

Summary

[0005] Provided herein are compounds having a structure of formula (I)



or an atropisomer thereof, a pharmaceutically acceptable salt thereof or a pharmaceutically acceptable salt of the atropisomer thereof.

[0006] The compounds disclosed herein can be in the form of a pharmaceutically acceptable salt. The compounds provided can be formulated into a pharmaceutical formulation comprising a compound disclosed herein and a pharmaceutically acceptable excipient.

[0007] Also provided is an *in vitro* method of inhibiting KRAS G12C in a cell, comprising contacting the cell with a compound or composition disclosed herein. Further provided is that the compounds disclosed herein are for use as a medicament, in the treatment of cancer in a subject comprising administering to the subject a therapeutically effective amount of a compound or composition disclosed herein. In some embodiments, the cancer is lung cancer, pancreatic cancer, or colorectal cancer.

Detailed description**Definitions**

[0008] **Abbreviations:** The following abbreviations may be used herein:

AcOH	acetic acid
aq or aq.	Aqueous
BOC or Boc	tert-butyloxycarbonyl

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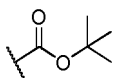
	cpme	cyclopentyl methyl ether
5	DCE	1,2-dichloroethane
	DABCO	1,4-diazabicyclo[2.2.2]octane
	DCM	Dichloromethane
	DMA	N,N-Dimethylacetamide
10	DMAP	4-dimethylaminopyridine
	DME	1,2-dimethoxyethane
	DMF	N,N-dimethylformamide
15	DMSO	dimethyl sulfoxide
	Dppf, DPPF or dppf	1,1'-bis(diphenylphosphino)ferrocene
	eq or eq. or equiv.	Equivalent
	ESI or ES	electrospray ionization
20	Et	Ethyl
	Et ₂ O	diethyl ether
	EtOAc	ethyl acetate
25	g	Grams
	h	Hour
	HPLC	high pressure liquid chromatography
	iPr	Isopropyl
30	iPr ₂ NEt or DIPEA	N-ethyl diisopropylamine (Hunig's base)
	KHMDS	potassium hexamethyldisilazide
	KOAc	potassium acetate
35	Lawesson's reagent	2,4-bis(4-methoxyphenyl)-2,4-dithioxo-1,3,2,4-dithiadiphosphetane, 2,4-Bis-(4-methoxyphenyl)-1,3-dithia-2,4-diphosphetane 2,4-disulfide
	LC MS, LCMS, LC-MS or LC/MS	liquid chromatography mass spectroscopy
40	LG	Leaving group (e.g., halogen, mesylate, triflate)
	LHMDS or LiHMDS	lithium hexamethyldisilazide
	m/z	mass divided by charge
	Me	Methyl
45	MeCN	Acetonitrile
	MeOH	Methanol
	Met	Metal species for cross-coupling (e.g., MgX, ZnX, SnR ₃ , SiR ₃ , B(OR) ₂)
50	mg	Milligrams
	min	Minutes
	mL	Milliliters
	MS	mass spectra
55	NaHMDS	sodium hexamethyldisilazide
	NBS	N-bromosuccinimide

(continued)

<i>n</i> -BuLi	<i>n</i> -butyllithium
NCS	N-chlorosuccinimide
NMR	nuclear magnetic resonance
Pd ₂ (dba) ₃	tris(dibenzylideneacetone)dipalladium(0)
Pd(dppf)Cl ₂ ·DCM	[1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium(II), complex with dichloromethane
Pd(PPh ₃) ₄	Tetrakis(triphenylphosphine)palladium(0)
Ph	Phenyl
PR or PG or Prot. group	protecting group
rbf	round-bottom flask
RP-HPLC	reverse phase high pressure liquid chromatography
RT or rt	room temperature
sat. or satd.	saturated
SFC	supercritical fluid chromatography
SPhos Pd G3 or SPhos G3	(2-Dicyclohexylphosphino-2',6'-dimethoxybiphenyl)[2-(2'-amino-1,1'-biphenyl)] palladium(II) methanesulfonate
TBAF	tetra- <i>n</i> -butylammonium fluoride
TBTU	<i>N,N,N',N'</i> - Tetramethyl- <i>O</i> -(benzotriazol-1-yl)uronium tetrafluoroborate
<i>t</i> -BuOH	<i>tert</i> -butanol
TEA or Et ₃ N	Trimethylamine
TFA	trifluoroacetic acid
THF	Tetrahydrofuran
UV	Ultraviolet

[0009] The use of the terms "a," "an," "the," and similar referents in the context of describing the invention (especially in the context of the claims) are to be construed to cover both the singular and the plural, unless otherwise indicated. Recitation of ranges of values herein merely are intended to serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein, and each separate value is incorporated into the specification as if it were individually recited herein. The use of any and all examples, or exemplary language (e.g., "such as") provided herein, is intended to better illustrate the invention and is not a limitation on the scope of the invention unless otherwise claimed. No language in the specification should be construed as indicating any non-claimed element as essential to the practice of the invention.

[0010] As used herein, the term Boc refers to the structure



Compounds of the disclosure

[0011] Provided herein are KRAS inhibitors having structure of Formula I, discussed in more detail below.

[0012] The compounds disclosed herein include all pharmaceutically acceptable isotopically-labeled compounds wherein one or more atoms of the compounds disclosed herein are replaced by atoms having the same atomic number, but an atomic mass or mass number different from the atomic mass or mass number usually found in nature. Examples of isotopes that can be incorporated into the disclosed compounds include isotopes of hydrogen, carbon, nitrogen, oxygen, phosphorous, fluorine, chlorine, and iodine, such as ²H, ³H, ¹¹C, ¹³C, ¹⁴C, ¹³N, ¹⁵N, ¹⁵O, ¹⁷O, ¹⁸O, ³¹P, ³²P,

³⁵S, ¹⁸F, ³⁶Cl, ¹²³I, and ¹²⁵I, respectively. These radiolabelled compounds could be useful to help determine or measure the effectiveness of the compounds, by characterizing, for example, the site or mode of action, or binding affinity to pharmacologically important site of action. Certain isotopically-labeled compounds of the disclosure, for example, those incorporating a radioactive isotope, are useful in drug and/or substrate tissue distribution studies. The radioactive isotopes tritium, i.e. ³H, and carbon-14, i.e. ¹⁴C, are particularly useful for this purpose in view of their ease of incorporation and ready means of detection.

[0013] Substitution with heavier isotopes such as deuterium, i.e. ²H, may afford certain therapeutic advantages resulting from greater metabolic stability, for example, increased in vivo half-life or reduced dosage requirements, and hence are preferred in some circumstances.

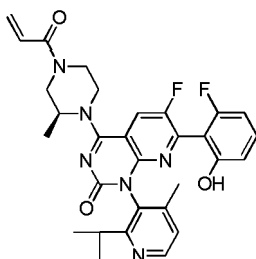
[0014] Substitution with positron emitting isotopes, such as ¹¹C, ¹⁸F, ¹⁵O and ¹³N, can be useful in Positron Emission Topography (PET) studies for examining substrate receptor occupancy. Isotopically-labeled compounds of structure (I) can generally be prepared by conventional techniques known to those skilled in the art or by processes analogous to those described in the Preparations and Examples as set out below using an appropriate isotopically-labeled reagent in place of the non-labeled reagent previously employed.

[0015] Isotopically-labeled compounds as disclosed herein can generally be prepared by conventional techniques known to those skilled in the art or by processes analogous to those described in the accompanying examples and schemes using an appropriate isotopically-labeled reagent in place of the non-labeled reagent previously employed.

[0016] The compounds disclosed herein include all tautomeric forms of the compounds.

[0017] Certain of the compounds disclosed herein may exist as atropisomers, which are conformational stereoisomers that occur when rotation about a single bond in the molecule is prevented, or greatly slowed, as a result of steric interactions with other parts of the molecule. The compounds disclosed herein include all atropisomers, both as pure individual atropisomer preparations, enriched preparations of each, or a non-specific mixture of each. Where the rotational barrier about the single bond is high enough, and interconversion between conformations is slow enough, separation and isolation of the isomeric species may be permitted.

[0018] The present invention discloses compounds having a structure selected from:



or an atropisomer thereof, a pharmaceutically acceptable salt thereof, or a pharmaceutically acceptable salt of the atropisomer thereof.

[0019] In another embodiment, these compounds can be used as intermediates in the process of making compounds in the present application.

[0020] In another embodiment, these compounds can be in the form of a pharmaceutically acceptable salt.

[0021] In another embodiment, these compounds can be in a pharmaceutical formulation comprising any one or more of the compounds and a pharmaceutically acceptable excipient.

[0022] In another embodiment, these compounds can be used in an *in vitro* method of inhibiting KRAS G12C in a cell, comprising contacting the cell with the compound of any one of compounds or the pharmaceutical formulation.

[0023] In another embodiment, these compounds are for use in the treatment of cancer in a subject comprising administering to the subject a therapeutically effective amount of any of the compounds or the compositions.

[0024] In another embodiment, the cancer is lung cancer, pancreatic cancer, or colorectal cancer.

[0025] In another embodiment, the cancer is lung cancer.

[0026] In another embodiment, the cancer is pancreatic cancer.

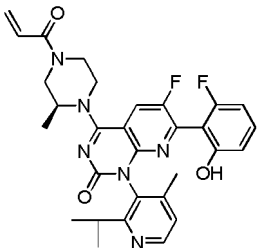
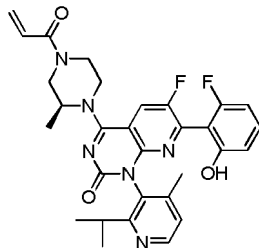
[0027] In another embodiment, the cancer is colorectal cancer.

[0028] In another embodiment, the present invention comprises the compounds for use in the treatment of cancer in a subject.

[0029] In another embodiment, the cancer is a hematologic malignancy.

[0030] Specifically contemplated compounds include those as listed in Table 1:

Table 1

Ex.#	Chemical Structure	Ex.#	Chemical Structure
1-1		1-2	
	1 st-eluting isomer		2nd-eluting isomer

Synthesis of disclosed compounds

[0031] Compounds as disclosed herein can be synthesized via a number of specific methods. The examples which outline specific synthetic routes, and the generic schemes below are meant to provide guidance to the ordinarily skilled synthetic chemist, who will readily appreciate that the solvent, concentration, reagent, protecting group, order of synthetic steps, time, temperature, and the like can be modified as necessary, well within the skill and judgment of the ordinarily skilled artisan.

Pharmaceutical compositions, dosing, and routes of administration

[0032] Also provided herein are pharmaceutical compositions that includes a compound as disclosed herein, together with a pharmaceutically acceptable excipient, such as, for example, a diluent or carrier. Compounds and pharmaceutical compositions suitable for use in the present invention include those wherein the compound can be administered in an effective amount to achieve its intended purpose. Administration of the compound described in more detail below.

[0033] Suitable pharmaceutical formulations can be determined by the skilled artisan depending on the route of administration and the desired dosage. See, e.g., Remington's Pharmaceutical Sciences, 1435-712 (18th ed., Mack Publishing Co, Easton, Pennsylvania, 1990). Formulations may influence the physical state, stability, rate of in vivo release and rate of in vivo clearance of the administered agents. Depending on the route of administration, a suitable dose may be calculated according to body weight, body surface areas or organ size. Further refinement of the calculations necessary to determine the appropriate treatment dose is routinely made by those of ordinary skill in the art without undue experimentation, especially in light of the dosage information and assays disclosed herein as well as the pharmacokinetic data obtainable through animal or human clinical trials.

[0034] The phrases "pharmaceutically acceptable" or "pharmacologically acceptable" refer to molecular entities and compositions that do not produce adverse, allergic, or other untoward reactions when administered to an animal or a human. As used herein, "pharmaceutically acceptable" includes any and all solvents, dispersion media, coatings, antibacterial and antifungal agents, isotonic and absorption delaying agents and the like. The use of such excipients for pharmaceutically active substances is well known in the art. Except insofar as any conventional media or agent is incompatible with the therapeutic compositions, its use in therapeutic compositions is contemplated. Supplementary active ingredients also can be incorporated into the compositions. In exemplary embodiments, the formulation may comprise corn syrup solids, high-oleic safflower oil, coconut oil, soy oil, L-leucine, calcium phosphate tribasic, L-tyrosine, L-proline, L-lysine acetate, DATEM (an emulsifier), L-glutamine, L-valine, potassium phosphate dibasic, L-isoleucine, L-arginine, L-alanine, glycine, L-asparagine monohydrate, L-serine, potassium citrate, L-threonine, sodium citrate, magnesium chloride, L-histidine, L-methionine, ascorbic acid, calcium carbonate, L-glutamic acid, L-cystine dihydrochloride, L-tryptophan, L-aspartic acid, choline chloride, taurine, m-inositol, ferrous sulfate, ascorbyl palmitate, zinc sulfate, L-carnitine, alpha-tocopheryl acetate, sodium chloride, niacinamide, mixed tocopherols, calcium pantothenate, cupric sulfate, thiamine chloride hydrochloride, vitamin A palmitate, manganese sulfate, riboflavin, pyridoxine hydrochloride, folic acid, beta-carotene, potassium iodide, phyloquinone, biotin, sodium selenate, chromium chloride, sodium molybdate, vitamin D3 and cyanocobalamin.

[0035] The compound can be present in a pharmaceutical composition as a pharmaceutically acceptable salt. As used herein, "pharmaceutically acceptable salts" include, for example base addition salts and acid addition salts.

[0036] Pharmaceutically acceptable base addition salts may be formed with metals or amines, such as alkali and alkaline earth metals or organic amines. Pharmaceutically acceptable salts of compounds may also be prepared with a pharmaceutically acceptable cation. Suitable pharmaceutically acceptable cations are well known to those skilled in the art and include alkaline, alkaline earth, ammonium and quaternary ammonium cations. Carbonates or hydrogen car-

bonates are also possible. Examples of metals used as cations are sodium, potassium, magnesium, ammonium, calcium, or ferric, and the like. Examples of suitable amines include isopropylamine, trimethylamine, histidine, N,N'-dibenzylethylenediamine, chlorprocaine, choline, diethanolamine, dicyclohexylamine, ethylenediamine, N-methylglucamine, and procaine.

[0037] Pharmaceutically acceptable acid addition salts include inorganic or organic acid salts. Examples of suitable acid salts include the hydrochlorides, formates, acetates, citrates, salicylates, nitrates, phosphates. Other suitable pharmaceutically acceptable salts are well known to those skilled in the art and include, for example, formic, acetic, citric, oxalic, tartaric, or mandelic acids, hydrochloric acid, hydrobromic acid, sulfuric acid or phosphoric acid; with organic carboxylic, sulfonic, sulfo or phospho acids or N-substituted sulfamic acids, for example acetic acid, trifluoroacetic acid (TFA), propionic acid, glycolic acid, succinic acid, maleic acid, hydroxymaleic acid, methylmaleic acid, fumaric acid, malic acid, tartaric acid, lactic acid, oxalic acid, gluconic acid, glucaric acid, glucuronic acid, citric acid, benzoic acid, cinnamic acid, mandelic acid, salicylic acid, 4-aminosalicylic acid, 2-phenoxybenzoic acid, 2-acetoxybenzoic acid, embonic acid, nicotinic acid or isonicotinic acid; and with amino acids, such as the 20 alpha amino acids involved in the synthesis of proteins in nature, for example glutamic acid or aspartic acid, and also with phenylacetic acid, methanesulfonic acid, ethanesulfonic acid, 2-hydroxyethanesulfonic acid, ethane 1,2-disulfonic acid, benzenesulfonic acid, 4-methylbenzenesulfonic acid, naphthalene 2-sulfonic acid, naphthalene 1,5-disulfonic acid, 2- or 3-phosphoglycerate, glucose 6-phosphate, N-cyclohexylsulfamic acid (with the formation of cyclamates), or with other acid organic compounds, such as ascorbic acid.

[0038] Pharmaceutical compositions containing the compounds disclosed herein can be manufactured in a conventional manner, e.g., by conventional mixing, dissolving, granulating, dragee-making, levigating, emulsifying, encapsulating, entrapping, or lyophilizing processes. Proper formulation is dependent upon the route of administration chosen.

[0039] For oral administration, suitable compositions can be formulated readily by combining a compound disclosed herein with pharmaceutically acceptable excipients such as carriers well known in the art. Such excipients and carriers enable the present compounds to be formulated as tablets, pills, dragees, capsules, liquids, gels, syrups, slurries, suspensions and the like, for oral ingestion by a patient to be treated. Pharmaceutical preparations for oral use can be obtained by adding a compound as disclosed herein with a solid excipient, optionally grinding a resulting mixture, and processing the mixture of granules, after adding suitable auxiliaries, if desired, to obtain tablets or dragee cores. Suitable excipients include, for example, fillers and cellulose preparations. If desired, disintegrating agents can be added. Pharmaceutically acceptable ingredients are well known for the various types of formulation and may be for example binders (e.g., natural or synthetic polymers), lubricants, surfactants, sweetening and flavoring agents, coating materials, preservatives, dyes, thickeners, adjuvants, antimicrobial agents, antioxidants and carriers for the various formulation types.

[0040] When a therapeutically effective amount of a compound disclosed herein is administered orally, the composition typically is in the form of a solid (e.g., tablet, capsule, pill, powder, or troche) or a liquid formulation (e.g., aqueous suspension, solution, elixir, or syrup).

[0041] When administered in tablet form, the composition can additionally contain a functional solid and/or solid carrier, such as a gelatin or an adjuvant. The tablet, capsule, and powder can contain about 1 to about 95% compound, and preferably from about 15 to about 90% compound.

[0042] When administered in liquid or suspension form, a functional liquid and/or a liquid carrier such as water, petroleum, or oils of animal or plant origin can be added. The liquid form of the composition can further contain physiological saline solution, sugar alcohol solutions, dextrose or other saccharide solutions, or glycols. When administered in liquid or suspension form, the composition can contain about 0.5 to about 90% by weight of a compound disclosed herein, and preferably about 1 to about 50% of a compound disclosed herein. In one embodiment contemplated, the liquid carrier is non-aqueous or substantially non-aqueous. For administration in liquid form, the composition may be supplied as a rapidly-dissolving solid formulation for dissolution or suspension immediately prior to administration.

[0043] When a therapeutically effective amount of a compound disclosed herein is administered by intravenous, cutaneous, or subcutaneous injection, the composition is in the form of a pyrogen-free, parenterally acceptable aqueous solution. The preparation of such parenterally acceptable solutions, having due regard to pH, isotonicity, stability, and the like, is within the skill in the art. A preferred composition for intravenous, cutaneous, or subcutaneous injection typically contains, in addition to a compound disclosed herein, an isotonic vehicle. Such compositions may be prepared for administration as solutions of free base or pharmacologically acceptable salts in water suitably mixed with a surfactant, such as hydroxypropylcellulose. Dispersions also can be prepared in glycerol, liquid polyethylene glycols, and mixtures thereof and in oils. Under ordinary conditions of storage and use, these preparations can optionally contain a preservative to prevent the growth of microorganisms.

[0044] Injectable compositions can include sterile aqueous solutions, suspensions, or dispersions and sterile powders for the extemporaneous preparation of sterile injectable solutions, suspensions, or dispersions. In all embodiments the form must be sterile and must be fluid to the extent that easy syringability exists. It must be stable under the conditions of manufacture and storage and must resist the contaminating action of microorganisms, such as bacteria and fungi, by optional inclusion of a preservative. The carrier can be a solvent or dispersion medium containing, for example, water,

ethanol, polyol (e.g., glycerol, propylene glycol, and liquid polyethylene glycol, and the like), suitable mixtures thereof, and vegetable oils. In one embodiment contemplated, the carrier is non-aqueous or substantially non-aqueous. The proper fluidity can be maintained, for example, by the use of a coating, such as lecithin, by the maintenance of the required particle size of the compound in the embodiment of dispersion and by the use of surfactants. The prevention of the action of microorganisms can be brought about by various antibacterial and antifungal agents, for example, parabens, chlorobutanol, phenol, sorbic acid, thimerosal, and the like. In many embodiments, it will be preferable to include isotonic agents, for example, sugars or sodium chloride. Prolonged absorption of the injectable compositions can be brought about by the use in the compositions of agents delaying absorption, for example, aluminum monostearate and gelatin.

[0045] Sterile injectable solutions are prepared by incorporating the active compounds in the required amount in the appropriate solvent with various of the other ingredients enumerated above, as required, followed by filtered sterilization. Generally, dispersions are prepared by incorporating the various sterilized active ingredients into a sterile vehicle which contains the basic dispersion medium and the required other ingredients from those enumerated above. In the embodiment of sterile powders for the preparation of sterile injectable solutions, the preferred methods of preparation are vacuum-drying and freeze-drying techniques which yield a powder of the active ingredient plus any additional desired ingredient from a previously sterile-filtered solution thereof.

[0046] Slow release or sustained release formulations may also be prepared in order to achieve a controlled release of the active compound in contact with the body fluids in the GI tract, and to provide a substantially constant and effective level of the active compound in the blood plasma. For example, release can be controlled by one or more of dissolution, diffusion, and ion-exchange. In addition, the slow release approach may enhance absorption via saturable or limiting pathways within the GI tract. For example, the compound may be embedded for this purpose in a polymer matrix of a biological degradable polymer, a water-soluble polymer or a mixture of both, and optionally suitable surfactants. Embedding can mean in this context the incorporation of micro-particles in a matrix of polymers. Controlled release formulations are also obtained through encapsulation of dispersed micro-particles or emulsified micro-droplets via known dispersion or emulsion coating technologies.

[0047] For administration by inhalation, compounds of the present invention are conveniently delivered in the form of an aerosol spray presentation from pressurized packs or a nebulizer, with the use of a suitable propellant. In the embodiment of a pressurized aerosol, the dosage unit can be determined by providing a valve to deliver a metered amount. Capsules and cartridges of, e.g., gelatin, for use in an inhaler or insufflator can be formulated containing a powder mix of the compound and a suitable powder base such as lactose or starch.

[0048] The compounds disclosed herein can be formulated for parenteral administration by injection (e.g., by bolus injection or continuous infusion). Formulations for injection can be presented in unit dosage form (e.g., in ampules or in multidose containers), with an added preservative. The compositions can take such forms as suspensions, solutions, or emulsions in oily or aqueous vehicles, and can contain formulatory agents such as suspending, stabilizing, and/or dispersing agents.

[0049] Pharmaceutical formulations for parenteral administration include aqueous solutions of the compounds in water-soluble form. Additionally, suspensions of the compounds can be prepared as appropriate oily injection suspensions. Suitable lipophilic solvents or vehicles include fatty oils or synthetic fatty acid esters. Aqueous injection suspensions can contain substances which increase the viscosity of the suspension. Optionally, the suspension also can contain suitable stabilizers or agents that increase the solubility of the compounds and allow for the preparation of highly concentrated solutions. Alternatively, a present composition can be in powder form for constitution with a suitable vehicle (e.g., sterile pyrogen-free water) before use.

[0050] Compounds disclosed herein also can be formulated in rectal compositions, such as suppositories or retention enemas (e.g., containing conventional suppository bases). In addition to the formulations described previously, the compounds also can be formulated as a depot preparation. Such long-acting formulations can be administered by implantation (e.g., subcutaneously or intramuscularly) or by intramuscular injection. Thus, for example, the compounds can be formulated with suitable polymeric or hydrophobic materials (for example, as an emulsion in an acceptable oil) or ion exchange resins, or as sparingly soluble derivatives, for example, as a sparingly soluble salt.

[0051] In particular, a compound disclosed herein can be administered orally, buccally, or sublingually in the form of tablets containing excipients, such as starch or lactose, or in capsules or ovules, either alone or in admixture with excipients, or in the form of elixirs or suspensions containing flavoring or coloring agents. Such liquid preparations can be prepared with pharmaceutically acceptable additives, such as suspending agents. A compound also can be injected parenterally, for example, intravenously, intramuscularly, subcutaneously, or intracoronarily. For parenteral administration, the compound is best used in the form of a sterile aqueous solution which can contain other substances, for example, salts, or sugar alcohols, such as mannitol, or glucose, to make the solution isotonic with blood.

[0052] For veterinary use, a compound disclosed herein is administered as a suitably acceptable formulation in accordance with normal veterinary practice. The veterinarian can readily determine the dosing regimen and route of administration that is most appropriate for a particular animal.

[0053] In some embodiments, all the necessary components for the treatment of KRAS-related disorder using a compound as disclosed herein either alone or in combination with another agent or intervention traditionally used for the treatment of such disease may be packaged into a kit. Specifically, the present invention provides a kit for use in the therapeutic intervention of the disease comprising a packaged set of medicaments that include the compound disclosed herein as well as buffers and other components for preparing deliverable forms of said medicaments, and/or devices for delivering such medicaments, and/or any agents that are used in combination therapy with the compound disclosed herein, and/or instructions for the treatment of the disease packaged with the medicaments. The instructions may be fixed in any tangible medium, such as printed paper, or a computer readable magnetic or optical medium, or instructions to reference a remote computer data source such as a world wide web page accessible via the internet.

[0054] A "therapeutically effective amount" means an amount effective to treat or to prevent development of, or to alleviate the existing symptoms of, the subject being treated. Determination of the effective amounts is well within the capability of those skilled in the art, especially in light of the detailed disclosure provided herein. Generally, a "therapeutically effective dose" refers to that amount of the compound that results in achieving the desired effect. For example, in one preferred embodiment, a therapeutically effective amount of a compound disclosed herein decreases KRAS activity by at least 5%, compared to control, at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, or at least 90%.

[0055] The amount of compound administered can be dependent on the subject being treated, on the subject's age, health, sex, and weight, the kind of concurrent treatment (if any), severity of the affliction, the nature of the effect desired, the manner and frequency of treatment, and the judgment of the prescribing physician. The frequency of dosing also can be dependent on pharmacodynamic effects on arterial oxygen pressures. However, the most preferred dosage can be tailored to the individual subject, as is understood and determinable by one of skill in the art, without undue experimentation. This typically involves adjustment of a standard dose (e.g., reduction of the dose if the patient has a low body weight).

[0056] While individual needs vary, determination of optimal ranges of effective amounts of the compound is within the skill of the art. For administration to a human in the curative or prophylactic treatment of the conditions and disorders identified herein, for example, typical dosages of the compounds of the present invention can be about 0.05 mg/kg/day to about 50 mg/kg/day, for example at least 0.05 mg/kg, at least 0.08 mg/kg, at least 0.1 mg/kg, at least 0.2 mg/kg, at least 0.3 mg/kg, at least 0.4 mg/kg, or at least 0.5 mg/kg, and preferably 50 mg/kg or less, 40 mg/kg or less, 30 mg/kg or less, 20 mg/kg or less, or 10 mg/kg or less, which can be about 2.5 mg/day (0.5 mg/kg x 5kg) to about 5000 mg/day (50mg/kg x 100kg), for example. For example, dosages of the compounds can be about 0.1 mg/kg/day to about 50 mg/kg/day, about 0.05 mg/kg/day to about 10 mg/kg/day, about 0.05 mg/kg/day to about 5 mg/kg/day, about 0.05 mg/kg/day to about 3 mg/kg/day, about 0.07 mg/kg/day to about 3 mg/kg/day, about 0.09 mg/kg/day to about 3 mg/kg/day, about 0.05 mg/kg/day to about 0.1 mg/kg/day, about 0.1 mg/kg/day to about 1 mg/kg/day, about 1 mg/kg/day to about 10 mg/kg/day, about 1 mg/kg/day to about 5 mg/kg/day, about 1 mg/kg/day to about 3 mg/kg/day, about 3 mg/day to about 500 mg/day, about 5 mg/day to about 250 mg/day, about 10 mg/day to about 100 mg/day, about 3 mg/day to about 10 mg/day, or about 100 mg/day to about 250 mg/day. Such doses may be administered in a single dose or it may be divided into multiple doses.

Methods of using KRAS G12C inhibitors

[0057] The present disclosure provides a method of inhibiting RAS-mediated cell signaling comprising contacting a cell with an effective amount of one or more compounds disclosed herein. Inhibition of RAS-mediated signal transduction can be assessed and demonstrated by a wide variety of ways known in the art. Non-limiting examples include a showing of (a) a decrease in GTPase activity of RAS; (b) a decrease in GTP binding affinity or an increase in GDP binding affinity; (c) an increase in K off of GTP or a decrease in K off of GDP; (d) a decrease in the levels of signaling transduction molecules downstream in the RAS pathway, such as a decrease in pMEK, pERK, or pAKT levels; and/or (e) a decrease in binding of RAS complex to downstream signaling molecules including but not limited to Raf. Kits and commercially available assays can be utilized for determining one or more of the above.

[0058] The disclosure also provides compounds or pharmaceutical compositions of the present disclosure for use in the treatment of disease conditions, including but not limited to conditions implicated by G12C KRAS, HRAS or NRAS mutation (e.g., cancer).

[0059] In some embodiments, an effective amount of any of the foregoing pharmaceutical compositions comprising a compound as disclosed herein for use in the treatment of cancer to a subject in need thereof. In some embodiments, the cancer is mediated by a KRAS, HRAS or NRAS G12C mutation. In various embodiments, the cancer is pancreatic cancer, colorectal cancer or lung cancer. In some embodiments, the cancer is gall bladder cancer, thyroid cancer, and bile duct cancer.

[0060] In some embodiments the disclosure provides compounds for use in the treatment of a disorder in a subject

in need thereof, wherein the said treatment comprises determining if the subject has a KRAS, HRAS or NRAS G12C mutation and if the subject is determined to have the KRAS, HRAS or NRAS G12C mutation, then administering to the subject a therapeutically effective dose of at least one compound as disclosed herein or a pharmaceutically acceptable salt thereof.

[0061] The disclosed compounds inhibit anchorage-independent cell growth and therefore have the potential to inhibit tumor metastasis. Accordingly, in another embodiment the disclosed compounds are provided for use in the treatment of inhibiting tumor metastasis, by comprising administering an effective amount a compound disclosed herein.

[0062] KRAS, HRAS or NRAS G12C mutations have also been identified in hematological malignancies (e.g., cancers that affect blood, bone marrow and/or lymph nodes). Accordingly, certain embodiments are directed to the disclosed compounds (e.g., in the form of a pharmaceutical composition) for use in the treatment of a hematological malignancy. Such malignancies include, but are not limited to leukemias and lymphomas. For example, the presently disclosed compounds can be used for treatment of diseases such as Acute lymphoblastic leukemia (ALL), Acute myelogenous leukemia (AML), Chronic lymphocytic leukemia (CLL), small lymphocytic lymphoma (SLL), Chronic myelogenous leukemia (CML), Acute monocytic leukemia (AMoL) and/ or other leukemias. In other embodiments, the compounds are useful for treatment of lymphomas such as all subtypes of Hodgkins lymphoma or non-Hodgkins lymphoma. In various embodiments, the compounds are for use in the treatment of plasma cell malignancies such as multiple myeloma, mantle cell lymphoma, and Waldenstrom's macroglobulinemia.

[0063] Determining whether a tumor or cancer comprises a G12C KRAS, HRAS or NRAS mutation can be undertaken by assessing the nucleotide sequence encoding the KRAS, HRAS or NRAS protein, by assessing the amino acid sequence of the KRAS, HRAS or NRAS protein, or by assessing the characteristics of a putative KRAS, HRAS or NRAS mutant protein. The sequence of wild-type human KRAS, HRAS or NRAS is known in the art, (e.g. Accession No. NP203524).

[0064] Methods for detecting a mutation in a KRAS, HRAS or NRAS nucleotide sequence are known by those of skill in the art. These methods include, but are not limited to, polymerase chain reaction-restriction fragment length polymorphism (PCR-RFLP) assays, polymerase chain reaction-single strand conformation polymorphism (PCR-SSCP) assays, real-time PCR assays, PCR sequencing, mutant allele-specific PCR amplification (MASA) assays, direct sequencing, primer extension reactions, electrophoresis, oligonucleotide ligation assays, hybridization assays, TaqMan assays, SNP genotyping assays, high resolution melting assays and microarray analyses. In some embodiments, samples are evaluated for G12C KRAS, HRAS or NRAS mutations by real-time PCR. In real-time PCR, fluorescent probes specific for the KRAS, HRAS or NRAS G12C mutation are used. When a mutation is present, the probe binds and fluorescence is detected. In some embodiments, the KRAS, HRAS or NRAS G12C mutation is identified using a direct sequencing method of specific regions (e.g., exon 2 and/or exon 3) in the KRAS, HRAS or NRAS gene. This technique will identify all possible mutations in the region sequenced.

[0065] Methods for detecting a mutation in a KRAS, HRAS or NRAS protein are known by those of skill in the art. These methods include, but are not limited to, detection of a KRAS, HRAS or NRAS mutant using a binding agent (e.g., an antibody) specific for the mutant protein, protein electrophoresis and Western blotting, and direct peptide sequencing.

[0066] Methods for determining whether a tumor or cancer comprises a G12C KRAS, HRAS or NRAS mutation can use a variety of samples. In some embodiments, the sample is taken from a subject having a tumor or cancer. In some embodiments, the sample is a fresh tumor/cancer sample. In some embodiments, the sample is a frozen tumor/cancer sample. In some embodiments, the sample is a formalin-fixed paraffin- embedded sample. In some embodiments, the sample is a circulating tumor cell (CTC) sample. In some embodiments, the sample is processed to a cell lysate. In some embodiments, the sample is processed to DNA or RNA.

[0067] The disclosure also relates to the disclosed compounds for use in the treatment of a hyperproliferative disorder in a mammal that comprises administering to said mammal a therapeutically effective amount of a compound as disclosed herein, or a pharmaceutically acceptable salt thereof. In some embodiments, said use relate to treatment of a subject who suffers from a cancer such as acute myeloid leukemia, cancer in adolescents, adrenocortical carcinoma childhood, AIDS-related cancers (e.g. Lymphoma and Kaposi's Sarcoma), anal cancer, appendix cancer, astrocytomas, atypical teratoid, basal cell carcinoma, bile duct cancer, bladder cancer, bone cancer, brain stem glioma, brain tumor, breast cancer, bronchial tumors, Burkitt lymphoma, carcinoid tumor, atypical teratoid, embryonal tumors, germ cell tumor, primary lymphoma, cervical cancer, childhood cancers, chordoma, cardiac tumors, chronic lymphocytic leukemia (CLL), chronic myelogenous leukemia (CML), chronic myeloproliferative disorders, colon cancer, colorectal cancer, craniopharyngioma, cutaneous T-cell lymphoma, extrahepatic ductal carcinoma in situ (DCIS), embryonal tumors, CNS cancer, endometrial cancer, ependymoma, esophageal cancer, esthesioneuroblastoma, ewing sarcoma, extracranial germ cell tumor, extragonadal germ cell tumor, eye cancer, fibrous histiocytoma of bone, gall bladder cancer, gastric cancer, gastrointestinal carcinoid tumor, gastrointestinal stromal tumors (GIST), germ cell tumor, gestational trophoblastic tumor, hairy cell leukemia, head and neck cancer, heart cancer, liver cancer, Hodgkin lymphoma, hypopharyngeal cancer, intraocular melanoma, islet cell tumors, pancreatic neuroendocrine tumors, kidney cancer, laryngeal cancer, lip and oral

cavity cancer, liver cancer, lobular carcinoma in situ (LCIS), lung cancer, lymphoma, metastatic squamous neck cancer with occult primary, midline tract carcinoma, mouth cancer, multiple endocrine neoplasia syndromes, multiple myeloma/plasma cell neoplasm, mycosis fungoides, myelodysplastic syndromes, myelodysplastic/myeloproliferative neoplasms, multiple myeloma, merkel cell carcinoma, malignant mesothelioma, malignant fibrous histiocytoma of bone and osteosarcoma, nasal cavity and paranasal sinus cancer, nasopharyngeal cancer, neuroblastoma, non-hodgkin lymphoma, non-small cell lung cancer (NSCLC), oral cancer, lip and oral cavity cancer, oropharyngeal cancer, ovarian cancer, pancreatic cancer, papillomatosis, paraganglioma, paranasal sinus and nasal cavity cancer, parathyroid cancer, penile cancer, pharyngeal cancer, pleuropulmonary blastoma, primary central nervous system (CNS) lymphoma, prostate cancer, rectal cancer, transitional cell cancer, retinoblastoma, rhabdomyosarcoma, salivary gland cancer, skin cancer, stomach (gastric) cancer, small cell lung cancer, small intestine cancer, soft tissue sarcoma, T-Cell lymphoma, testicular cancer, throat cancer, thymoma and thymic carcinoma, thyroid cancer, transitional cell cancer of the renal pelvis and ureter, trophoblastic tumor, unusual cancers of childhood, urethral cancer, uterine sarcoma, vaginal cancer, vulvar cancer, or viral-induced cancer. In some embodiments, said treatment relates to a non-cancerous hyperproliferative disorder such as benign hyperplasia of the skin (e. g., psoriasis), restenosis, or prostate (e. g., benign prostatic hypertrophy (BPH)).

[0068] In some embodiments, the treatment is directed to treating lung cancers, comprising administering an effective amount of any of the above described compound (or a pharmaceutical composition comprising the same) to a subject in need thereof. In certain embodiments the lung cancer is a non- small cell lung carcinoma (NSCLC), for example adenocarcinoma, squamous-cell lung carcinoma or large-cell lung carcinoma. In some embodiments, the lung cancer is a small cell lung carcinoma. Other lung cancers treatable with the disclosed compounds include, but are not limited to, glandular tumors, carcinoid tumors and undifferentiated carcinomas.

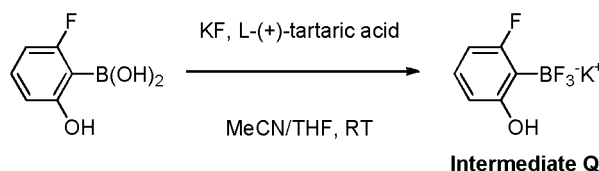
[0069] The disclosure further provides methods of modulating a G12C Mutant KRAS, HRAS or NRAS protein activity by contacting the protein with an effective amount of a compound of the disclosure. Modulation can be inhibiting or activating protein activity. In some embodiments, the disclosure provides methods of inhibiting protein activity by contacting the G12C Mutant KRAS, HRAS or NRAS protein with an effective amount of a compound of the disclosure in solution. In some embodiments, the disclosure provides methods of inhibiting the G12C Mutant KRAS, HRAS or NRAS protein activity by contacting a cell, tissue, or organ that expresses the protein of interest. In some embodiments, the disclosure provides methods of inhibiting protein activity in subject including but not limited to rodents and mammal (e.g., human) by administering into the subject an effective amount of a compound of the disclosure. In some embodiments, the percentage modulation exceeds 25%, 30%, 40%, 50%, 60%, 70%, 80%, or 90%. In some embodiments, the percentage of inhibiting exceeds 25%, 30%, 40%, 50%, 60%, 70%, 80%, or 90%.

[0070] In some embodiments, the disclosure provides methods of inhibiting KRAS, HRAS or NRAS G12C activity in a cell by contacting said cell with an amount of a compound of the disclosure sufficient to inhibit the activity of KRAS, HRAS or NRAS G12C in said cell. In some embodiments, the disclosure provides methods of inhibiting KRAS, HRAS or NRAS G12C activity in a tissue by contacting said tissue with an amount of a compound of the disclosure sufficient to inhibit the activity of KRAS, HRAS or NRAS G12C in said tissue. In some embodiments, the disclosure provides methods of inhibiting KRAS, HRAS or NRAS G12C activity in an organism by contacting said organism with an amount of a compound of the disclosure sufficient to inhibit the activity of KRAS, HRAS or NRAS G12C in said organism. In some embodiments, the disclosure provides methods of inhibiting KRAS, HRAS or NRAS G12C activity in an animal by contacting said animal with an amount of a compound of the disclosure sufficient to inhibit the activity of KRAS, HRAS or NRAS G12C in said animal. In some embodiments, the disclosure provides methods of inhibiting KRAS, HRAS or NRAS G12C activity in a mammal by contacting said mammal with an amount of a compound of the disclosure sufficient to inhibit the activity of KRAS, HRAS or NRAS G12C in said mammal. In some embodiments, the disclosure provides methods of inhibiting KRAS, HRAS or NRAS G12C activity in a human by contacting said human with an amount of a compound of the disclosure sufficient to inhibit the activity of KRAS, HRAS or NRAS G12C in said human. The present disclosure provides methods of treating a disease mediated by KRAS, HRAS or NRAS G12C activity in a subject in need of such treatment.

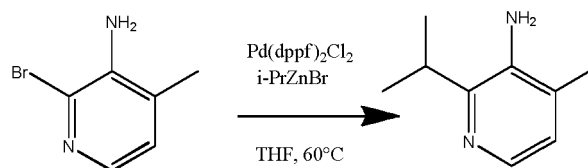
Examples

Preparation of intermediates Q and R:

[0071]



[0072] (2-Fluoro-6-hydroxyphenyl)potassium trifluoroborate (Intermediate Q). A solution of potassium fluoride (44.7 g, 770 mmol) in water (75 mL) was added to a suspension of (2-fluoro-6-hydroxyphenyl)boronic acid (30 g, 192 mmol, Combi-Blocks, San Diego, CA, USA) in acetonitrile (750 mL). The mixture was stirred for 2 min and then a solution of L-(+)-tartaric acid (72.2 g, 481 mmol) in THF (375 mL) was added over a 10 min period via addition funnel. The mixture was stirred vigorously with a mechanical stirrer for 1 h, and then the resulting suspension was filtered, and the filtered solids were washed with a small amount of THF. The solids were discarded and the filtrate was partially concentrated until solids started to precipitate out of solution. The mixture was then cooled to -20 °C and stirred for 16 h. The reaction was slowly warmed and 2-propanol (20 mL) was added. The resulting suspension was filtered and the filtered solids were washed with 2-propanol. The filtrate was again partially concentrated until a suspension formed and then was cooled to -20 °C and stirred for an additional 20 min. The resulting suspension was diluted with 2-propanol and filtered, and the filtered solids were washed with 2-propanol. The two batches of solids were combined to provide 2-fluoro-6-hydroxyphenyl)potassium trifluoroborate (**Intermediate Q**). ¹H NMR (400 MHz, DMSO-*d*₆) δ ppm 8.07 (q, *J* = 14.7 Hz, 1 H) 6.93 (q, *J* = 7.5 Hz, 1 H) 6.30-6.38 (m, 2 H).

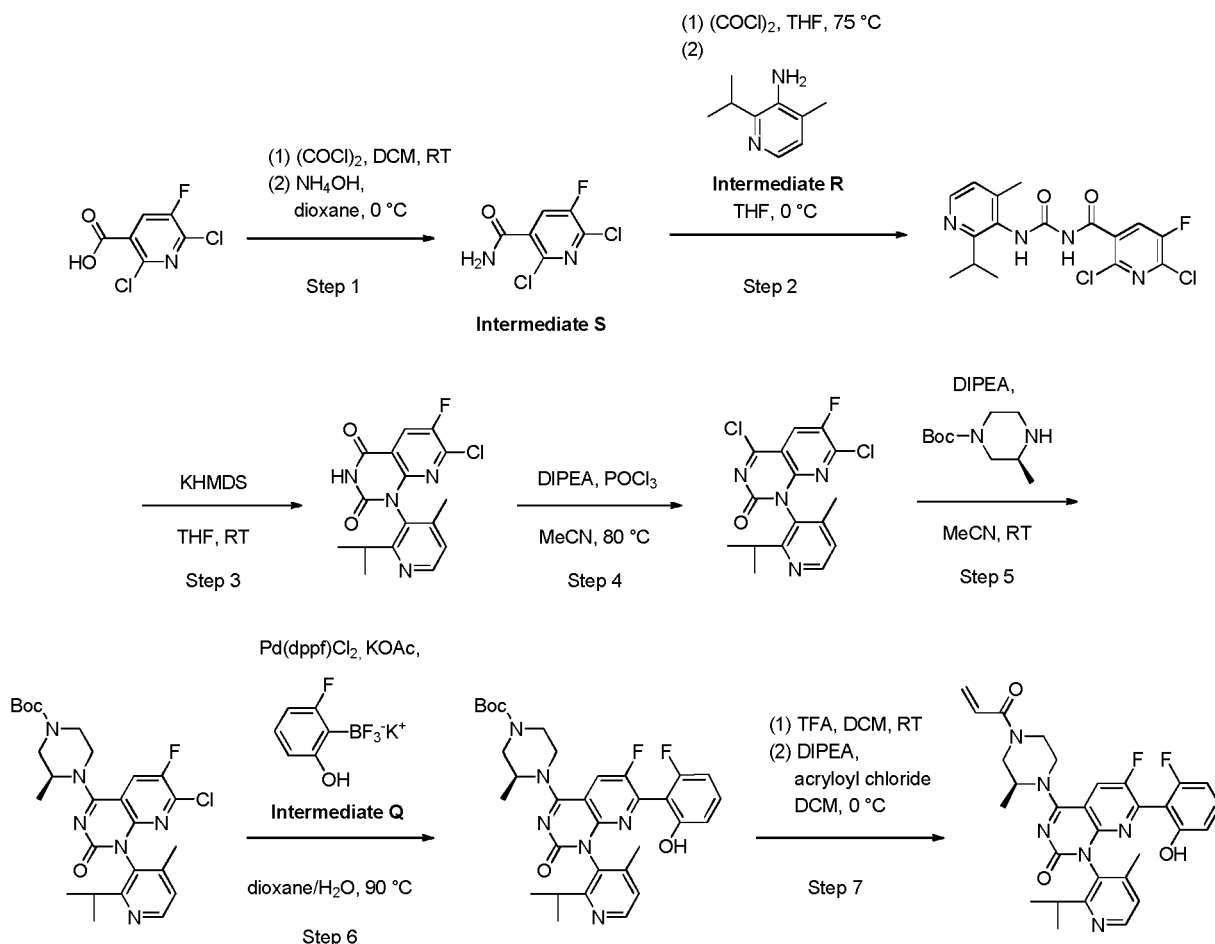


[0073] 2-Isopropyl-4-methylpyridin-3-amine (Intermediate R). To a slurry of 3-amino-2-bromo-4-picoline (360 mg, 1.9 mmol, Combi-Blocks, San Diego, CA, USA) in THF (4 mL) was added [1,1'-bis(diphenylphosphino)ferrocene]dichloropalladium(II), complex with dichloromethane (79 mg, 0.10 mmol). The resulting slurry was deoxygenated with argon for 2 min and then 2-propylzinc bromide (0.5 M solution in THF, 5.40 mL, 2.7 mmol, Sigma-Aldrich, St. Louis, MO) was added. The resulting solution was heated at 60 °C for 17 h, then the heating was stopped and the reaction was allowed to cool to room temperature. The reaction mixture was quenched with water (10 mL) and 1 N NaOH solution (20 mL) and then was extracted with EtOAc (2x). The combined organic layers were dried over anhydrous sodium sulfate and concentrated. The residue was purified by silica gel chromatography (eluent: 0-15% MeOH/DCM) to provide 2-isopropyl-4-methylpyridin-3-amine. ¹H NMR (400 MHz, DMSO-*d*₆) δ ppm 7.66 (d, *J*=4.6 Hz, 1 H), 6.78 (d, *J*=4.8 Hz, 1 H), 4.72 (br s, 2 H), 3.14-3.25 (m, 1 H), 2.08 (s, 3 H), 1.14 (d, *J*= 6.8 Hz, 6 H). *m/z* (ESI, +ve ion): 151.1 (M+H)⁺.

Example 1

6-Fluoro-7-(2-fluoro-6-hydroxyphenyl)-1-(4-methyl-2-(2-propanyl)-3-pyridinyl)-4-((2S)-2-methyl-4-(2-propenoyl)-1-piperazinyl)pyrido[2,3-*d*]pyrimidin-2(1H)-one

[0074]



[0075] Step 1: 2,6-Dichloro-5-fluoronicotinamide (Intermediate S). To a mixture of 2,6-dichloro-5-fluoro-nicotinic acid (4.0 g, 19.1 mmol, AstaTech Inc., Bristol, PA) in dichloromethane (48 mL) was added oxalyl chloride (2M solution in DCM, 11.9 mL, 23.8 mmol), followed by a catalytic amount of DMF (0.05 mL). The reaction was stirred at room temperature overnight and then was concentrated. The residue was dissolved in 1,4-dioxane (48 mL) and cooled to 0 °C. Ammonium hydroxide solution (28.0-30% NH₃ basis, 3.6 mL, 28.6 mmol) was added slowly via syringe. The resulting mixture was stirred at 0 °C for 30 min and then was concentrated. The residue was diluted with a 1:1 mixture of EtOAc/Heptane and agitated for 5 min, then was filtered. The filtered solids were discarded, and the remaining mother liquor was partially concentrated to half volume and filtered. The filtered solids were washed with heptane and dried in a reduced-pressure oven (45 °C) overnight to provide 2,6-dichloro-5-fluoronicotinamide. ¹HNMR (400 MHz, DMSO-*d*₆) δ ppm 8.23 (d, *J*=7.9 Hz, 1 H) 8.09 (br s, 1 H) 7.93 (br s, 1 H). *m/z* (ESI, +ve ion): 210.9 (M+H)⁺.

[0076] Step 2: 2,6-Dichloro-5-fluoro-N-((2-isopropyl-4-methylpyridin-3-yl)carbamoyl)nicotinamide. To an ice-cooled slurry of 2,6-dichloro-5-fluoronicotinamide (Intermediate S, 5.0 g, 23.9 mmol) in THF (20 mL) was added oxalyl chloride (2 M solution in DCM, 14.4 mL, 28.8 mmol) slowly via syringe. The resulting mixture was heated at 75 °C for 1 h, then heating was stopped, and the reaction was concentrated to half volume. After cooling to 0 °C, THF (20 mL) was added, followed by a solution of 2-isopropyl-4-methylpyridin-3-amine (Intermediate R, 3.59 g, 23.92 mmol) in THF (10 mL), dropwise via cannula. The resulting mixture was stirred at 0 °C for 1 h and then was quenched with a 1:1 mixture of brine and saturated aqueous ammonium chloride. The mixture was extracted with EtOAc (3x) and the combined organic layers were dried over anhydrous sodium sulfate and concentrated to provide 2,6-dichloro-5-fluoro-N-((2-isopropyl-4-methylpyridin-3-yl)carbamoyl)nicotinamide. This material was used without further purification in the following step. *m/z* (ESI, +ve ion): 385.1 (M+H)⁺.

[0077] Step 3: 7-Chloro-6-fluoro-1-(2-isopropyl-4-methylpyridin-3-yl)pyrido[2,3-*d*]pyrimidine-2,4(1H,3H)-di-one. To an ice-cooled solution of 2,6-dichloro-5-fluoro-N-((2-isopropyl-4-methylpyridin-3-yl)carbamoyl)nicotinamide (9.2 g, 24.0 mmol) in THF (40 mL) was added KHMDS (1 M solution in THF, 50.2 mL, 50.2 mmol) slowly via syringe. The ice bath was removed and the resulting mixture was stirred for 40 min at room temperature. The reaction was quenched with saturated aqueous ammonium chloride and extracted with EtOAc (3x). The combined organic layers were dried over anhydrous sodium sulfate and concentrated. The residue was purified by silica gel chromatography (eluent: 0-50% 3:1 EtOAc-EtOH/heptane) to provide 7-chloro-6-fluoro-1-(2-isopropyl-4-methylpyridin-3-yl)pyrido[2,3-*d*]pyrimidine-

2,4(1H,3H)-dione. ^1H NMR (400 MHz, $\text{DMSO}-d_6$) δ ppm 12.27 (br s, 1H), 8.48-8.55 (m, 2H), 7.29 (d, $J=4.8$ Hz, 1H), 2.87 (quin, $J=6.6$ Hz, 1H), 1.99-2.06 (m, 3H), 1.09 (d, $J=6.6$ Hz, 3H), 1.01 (d, $J=6.6$ Hz, 3H). ^{19}F NMR (376 MHz, $\text{DMSO}-d_6$) δ : -126.90 (s, 1F). m/z (ESI, +ve ion): 349.1 (M+H) $^+$.

[0078] Step 4: 4,7-Dichloro-6-fluoro-1-(2-isopropyl-4-methylpyridin-3-yl)pyrido[2,3-*d*]pyrimidin-2(1H)-one. To a solution of 7-chloro-6-fluoro-1-(2-isopropyl-4-methylpyridin-3-yl)pyrido[2,3-*d*]pyrimidine-2,4(1H,3H)-dione (4.7 g, 13.5 mmol) and DIPEA (3.5 mL, 20.2 mmol) in acetonitrile (20 mL) was added phosphorus oxychloride (1.63 mL, 17.5 mmol), dropwise via syringe. The resulting mixture was heated at 80 °C for 1 h, and then was cooled to room temperature and concentrated to provide 4,7-dichloro-6-fluoro-1-(2-isopropyl-4-methylpyridin-3-yl)pyrido[2,3-*d*]pyrimidin-2(1H)-one. This material was used without further purification in the following step. m/z (ESI, +ve ion): 367.1 (M+H) $^+$.

[0079] Step 5: (S)-*tert*-Butyl 4-(7-chloro-6-fluoro-1-(2-isopropyl-4-methylpyridin-3-yl)-2-oxo-1,2-dihydropyrido[2,3-*d*]pyrimidin-4-yl)-3-methylpiperazine-1-carboxylate. To an ice-cooled solution of 4,7-dichloro-6-fluoro-1-(2-isopropyl-4-methylpyridin-3-yl)pyrido[2,3-*d*]pyrimidin-2(1H)-one (13.5 mmol) in acetonitrile (20 mL) was added DIPEA (7.1 mL, 40.3 mmol), followed by (S)-4-*N*-Boc-2-methyl piperazine (3.23 g, 16.1 mmol, Combi-Blocks, Inc., San Diego, CA, USA). The resulting mixture was warmed to room temperature and stirred for 1 h, then was diluted with cold saturated aqueous sodium bicarbonate solution (200 mL) and EtOAc (300 mL). The mixture was stirred for an additional 5 min, the layers were separated, and the aqueous layer was extracted with more EtOAc (1x). The combined organic layers were dried over anhydrous sodium sulfate and concentrated. The residue was purified by silica gel chromatography (eluent: 0-50% EtOAc/heptane) to provide (S)-*tert*-butyl 4-(7-chloro-6-fluoro-1-(2-isopropyl-4-methylpyridin-3-yl)-2-oxo-1,2-dihydropyrido[2,3-*d*]pyrimidin-4-yl)-3-methylpiperazine-1-carboxylate. m/z (ESI, +ve ion): 531.2 (M+H) $^+$.

[0080] Step 6: (3S)-*tert*-Butyl 4-(6-fluoro-7-(2-fluoro-6-hydroxyphenyl)-1-(2-isopropyl-4-methylpyridin-3-yl)-2-oxo-1,2-dihydropyrido[2,3-*d*]pyrimidin-4-yl)-3-methylpiperazine-1-carboxylate. A mixture of (S)-*tert*-butyl 4-(7-chloro-6-fluoro-1-(2-isopropyl-4-methylpyridin-3-yl)-2-oxo-1,2-dihydropyrido[2,3-*d*]pyrimidin-4-yl)-3-methylpiperazine-1-carboxylate (4.3 g, 8.1 mmol), potassium trifluoro(2-fluoro-6-hydroxyphenyl)borate (**Intermediate Q**, 2.9 g, 10.5 mmol), potassium acetate (3.2 g, 32.4 mmol) and [1,1'-bis(diphenylphosphino)ferrocene]dichloropalladium(II), complex with dichloromethane (661 mg, 0.81 mmol) in 1,4-dioxane (80 mL) was degassed with nitrogen for 1 min. De-oxygenated water (14 mL) was added, and the resulting mixture was heated at 90 °C for 1 h. The reaction was allowed to cool to room temperature, quenched with half-saturated aqueous sodium bicarbonate, and extracted with EtOAc (2x) and DCM (1x). The combined organic layers were dried over anhydrous sodium sulfate and concentrated. The residue was purified by silica gel chromatography (eluent: 0-60% 3:1 EtOAc-EtOH/heptane) to provide (3S)-*tert*-butyl 4-(6-fluoro-7-(2-fluoro-6-hydroxyphenyl)-1-(2-isopropyl-4-methylpyridin-3-yl)-2-oxo-1,2-dihydropyrido[2,3-*d*]pyrimidin-4-yl)-3-methylpiperazine-1-carboxylate. ^1H NMR (400 MHz, $\text{DMSO}-d_6$) δ ppm 10.19 (br s, 1H), 8.38 (d, $J=5.0$ Hz, 1H), 8.26 (dd, $J=12.5, 9.2$ Hz, 1H), 7.23-7.28 (m, 1H), 7.18 (d, $J=5.0$ Hz, 1H), 6.72 (d, $J=8.0$ Hz, 1H), 6.68 (t, $J=8.9$ Hz, 1H), 4.77-4.98 (m, 1H), 4.24 (br t, $J=14.2$ Hz, 1H), 3.93-4.08 (m, 1H), 3.84 (br d, $J=12.9$ Hz, 1H), 3.52-3.75 (m, 1H), 3.07-3.28 (m, 1H), 2.62-2.74 (m, 1H), 1.86-1.93 (m, 3H), 1.43-1.48 (m, 9H), 1.35 (dd, $J=10.8, 6.8$ Hz, 3H), 1.26-1.32 (m, 1H), 1.07 (dd, $J=6.6, 1.7$ Hz, 3H), 0.93 (dd, $J=6.6, 2.1$ Hz, 3H). ^{19}F NMR (376 MHz, $\text{DMSO}-d_6$) δ : -115.65 (s, 1F), -128.62 (s, 1F). m/z (ESI, +ve ion): 607.3 (M+H) $^+$.

[0081] Step 7: 6-Fluoro-7-(2-fluoro-6-hydroxyphenyl)-1-(4-methyl-2-(2-propenyl)-3-pyridinyl)-4-((2S)-2-methyl-4-(2-propenoyl)-1-piperazinyl)pyrido[2,3-*d*]pyrimidin-2(1H)-one. Trifluoroacetic acid (25 mL, 324 mmol) was added to a solution of (3S)-*tert*-butyl 4-(6-fluoro-7-(2-fluoro-6-hydroxyphenyl)-1-(2-isopropyl-4-methylpyridin-3-yl)-2-oxo-1,2-dihydropyrido[2,3-*d*]pyrimidin-4-yl)-3-methylpiperazine-1-carboxylate (6.3 g, 10.4 mmol) in DCM (30 mL). The resulting mixture was stirred at room temperature for 1 h and then was concentrated. The residue was dissolved in DCM (30 mL), cooled to 0 °C, and sequentially treated with DIPEA (7.3 mL, 41.7 mmol) and a solution of acryloyl chloride (0.849 mL, 10.4 mmol) in DCM (3 mL; added dropwise via syringe). The reaction was stirred at 0 °C for 10 min, then was quenched with half-saturated aqueous sodium bicarbonate and extracted with DCM (2x). The combined organic layers were dried over anhydrous sodium sulfate and concentrated. The residue was purified by silica gel chromatography (eluent: 0-100% 3:1 EtOAc-EtOH/heptane) to provide 6-fluoro-7-(2-fluoro-6-hydroxyphenyl)-1-(4-methyl-2-(2-propenyl)-3-pyridinyl)-4-((2S)-2-methyl-4-(2-propenoyl)-1-piperazinyl)pyrido[2,3-*d*]pyrimidin-2(1H)-one. ^1H NMR (400 MHz, $\text{DMSO}-d_6$) δ ppm 10.20 (s, 1H), 8.39 (d, $J=4.8$ Hz, 1H), 8.24-8.34 (m, 1H), 7.23-7.32 (m, 1H), 7.19 (d, $J=5.0$ Hz, 1H), 6.87 (td, $J=16.3, 11.0$ Hz, 1H), 6.74 (d, $J=8.6$ Hz, 1H), 6.69 (t, $J=8.6$ Hz, 1H), 6.21 (br d, $J=16.2$ Hz, 1H), 5.74-5.80 (m, 1H), 4.91 (br s, 1H), 4.23-4.45 (m, 2H), 3.97-4.21 (m, 1H), 3.44-3.79 (m, 2H), 3.11-3.31 (m, 1H), 2.67-2.77 (m, 1H), 1.91 (s, 3H), 1.35 (d, $J=6.8$ Hz, 3H), 1.08 (d, $J=6.6$ Hz, 3H), 0.94 (d, $J=6.8$ Hz, 3H). ^{19}F NMR (376 MHz, $\text{DMSO}-d_6$) δ ppm -115.64 (s, 1F), -128.63 (s, 1F). m/z (ESI, +ve ion): 561.2 (M+H) $^+$.

Table 2: Separated Compound Examples Including Stereoisomers, some of which are Atropisomers

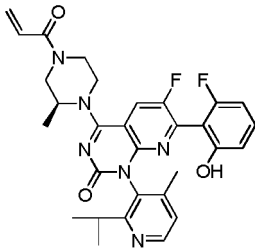
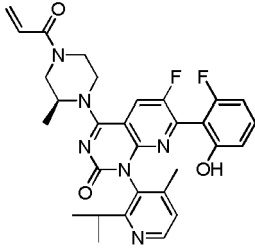
Ex.#	Chemical Structure	Name	Racemic SM / separation conditions
1-1	 1st-eluting isomer	6-fluoro-7-(2-fluoro-6-hydroxyphenyl)-1-(4-methyl-2-(2-propenyl)-3-pyridinyl)-4-((2S)-2-methyl-4-(2-propenoyl)-1-piperazinyl)pyrido[2,3-d]pyrimidin-2(1H)-one	1/ SFC (Chiralpak IC, 30 x 250 mm, 5 μ m, 55% MeOH/CO ₂ , 120 g/min, 10.3 MPa (103 bar)).
1-2	 2nd-eluting isomer	6-fluoro-7-(2-fluoro-6-hydroxyphenyl)-1-(4-methyl-2-(2-propenyl)-3-pyridinyl)-4-((2S)-2-methyl-4-(2-propenoyl)-1-piperazinyl)pyrido[2,3-d]pyrimidin-2(1H)-one	1/ SFC (Chiralpak IC, 30 x 250 mm, 5 μ m, 55% MeOH/CO ₂ , 120 g/min, 10.3 MPa (103 bar)).

Table 3: Analytical Data for Individual Examples

1-1	561.2	¹ H NMR (400 MHz, DMSO- <i>d</i> ₆) δ ppm 10.20 (s, 1 H), 8.39 (d, <i>J</i> = 4.8 Hz, 1 H), 8.24-8.34 (m, 1 H), 7.23-7.32 (m, 1 H), 7.19 (d, <i>J</i> = 5.0 Hz, 1 H), 6.87 (td, <i>J</i> = 16.3, 11.0 Hz, 1 H), 6.74 (d, <i>J</i> = 8.6 Hz, 1 H), 6.69 (t, <i>J</i> = 8.6 Hz, 1 H), 6.21 (br d, <i>J</i> = 16.2 Hz, 1 H), 5.74-5.80 (m, 1 H), 4.91 (br s, 1 H), 4.23-4.45 (m, 2 H), 3.97-4.21 (m, 1 H), 3.44-3.79 (m, 2 H), 3.11-3.31 (m, 1 H), 2.67-2.77 (m, 1 H), 1.91 (s, 3 H), 1.35 (d, <i>J</i> = 6.8 Hz, 3 H), 1.08 (d, <i>J</i> = 6.6 Hz, 3 H), 0.94 (d, <i>J</i> = 6.8 Hz, 3 H).
1-2	561.2	¹ H NMR (400 MHz, DMSO- <i>d</i> ₆) δ ppm 10.20 (s, 1 H), 8.39 (d, <i>J</i> = 4.8 Hz, 1 H), 8.24-8.34 (m, 1 H), 7.23-7.32 (m, 1 H), 7.19 (d, <i>J</i> = 5.0 Hz, 1 H), 6.87 (td, <i>J</i> = 16.3, 11.0 Hz, 1 H), 6.74 (d, <i>J</i> = 8.6 Hz, 1 H), 6.69 (t, <i>J</i> = 8.6 Hz, 1 H), 6.21 (br d, <i>J</i> = 16.2 Hz, 1 H), 5.74-5.80 (m, 1 H), 4.91-4.99 (br s, 1 H), 4.23-4.45 (m, 2 H), 3.97-4.21 (m, 1 H), 3.44-3.79 (m, 2 H), 3.11-3.31 (m, 1 H), 2.67-2.77 (m, 1 H), 1.91 (s, 3 H), 1.35 (d, <i>J</i> = 6.8 Hz, 3 H), 1.08 (d, <i>J</i> = 6.6 Hz, 3 H), 0.94 (d, <i>J</i> = 6.8 Hz, 3 H).

Biological assay data

[0082] Coupled Nucleotide Exchange Assay: Purified GDP-bound KRAS protein (aa 1-169), containing both G12C and C118A amino acid substitutions and an *N*-terminal His-tag, was pre-incubated with a compound dose-response titration for 5 min in assay buffer (25 mM HEPES pH 7.4, 10 mM MgCl₂, and 0.01% Triton X-100). Following compound preincubation, purified SOS protein (aa 564-1049) and GTP (Roche 10106399001) were added to the assay wells and incubated for an additional 30 min. To determine the extent of inhibition of SOS-mediated nucleotide exchange, purified GST-tagged cRAF (aa 1-149), nickel chelate AlphaLISA acceptor beads (PerkinElmer AL108R), and AlphaScreen glutathione donor beads (PerkinElmer 6765302) were added to the assay wells and incubated for 5 minutes. The assay plates were then read on a PerkinElmer EnVision Multilabel Reader, using AlphaScreen® technology, and data were analyzed using a 4-parameter logistic model to calculate IC₅₀ values.

[0083] Phospho-ERK1/2 MSD Assay: MIA PaCa-2 (ATCC® CRL-1420™) and A549 (ATCC® CCL-185™) cells were cultured in RPMI 1640 Medium (ThermoFisher Scientific 11875093) containing 10% fetal bovine serum (ThermoFisher Scientific 16000044) and 1x penicillin-streptomycin-glutamine (ThermoFisher Scientific 10378016). Sixteen hours prior to compound treatment, MIA PaCa-2 or A549 cells were seeded in 96-well cell culture plates at a density of 25,000 cells/well and incubated at 37°C, 5% CO₂. A compound dose-response titration was diluted in growth media, added to appropriate wells of a cell culture plate, and then incubated at 37°C, 5% CO₂ for 2 hours. Following compound treatment,

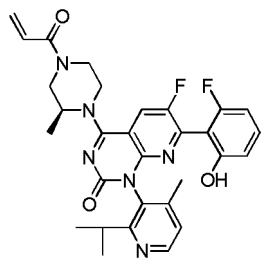
cells were stimulated with 10 ng/mL EGF (Roche 11376454001) for 10 min, washed with ice-cold Dulbecco's phosphate-buffered saline, no Ca^{2+} or Mg^{2+} (ThermoFisher Scientific 14190144), and then lysed in RIPA buffer (50 mM Tris-HCl pH 7.5, 1% Igepal, 0.5% sodium deoxycholate, 150 mM NaCl, and 0.5% sodium dodecyl sulfate) containing protease inhibitors (Roche 4693132001) and phosphatase inhibitors (Roche 4906837001). Phosphorylation of ERK1/2 in compound-treated lysates was assayed using Phospho-ERK1/2 Whole Cell Lysate kits (Meso Scale Discovery K151DWD) according to the manufacturer's protocol. Assay plates were read on a Meso Scale Discovery Sector Imager 6000, and data were analyzed using a 4-parameter logistic model to calculate IC_{50} values.

Table 4: Biochemical and cellular activity of compounds

Ex.#	Coupled exchange IC_{50} (μM)	p-ERK IC_{50} (MIA PaCa-2, μM)	p-ERK IC_{50} (A549, μM)
1	0.136	0.085	-
1-1	0.093	0.072	-
1-2	1.39	1.66	-

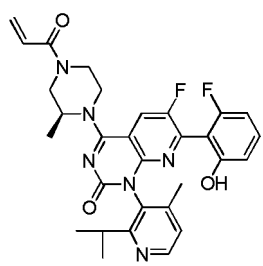
Claims

1. 6-Fluoro-7-(2-fluoro-6-hydroxyphenyl)-1-(4-methyl-2-(2-propenyl)-3-pyridinyl)-4-((2S)-2-methyl-4-(2-propenoyl)-1-piperazinyl)pyrido[2,3-d]pyrimidin-2(1H)-one having the following chemical structure



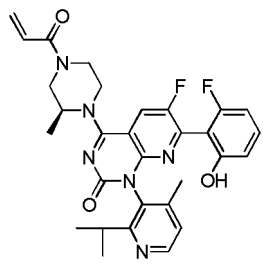
or an atropisomer thereof, a pharmaceutically acceptable salt thereof, or a pharmaceutically acceptable salt of the atropisomer thereof.

2. 6-Fluoro-7-(2-fluoro-6-hydroxyphenyl)-1-(4-methyl-2-(2-propenyl)-3-pyridinyl)-4-((2S)-2-methyl-4-(2-propenoyl)-1-piperazinyl)pyrido[2,3-d]pyrimidin-2(1 H)-one, according to claim 1, having the following chemical structure



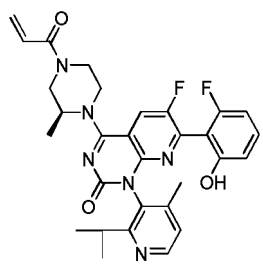
or an atropisomer thereof, or a pharmaceutically acceptable salt of the atropisomer thereof.

3. Atropisomer 1 of 6-fluoro-7-(2-fluoro-6-hydroxyphenyl)-1-(4-methyl-2-(2-propenyl)-3-pyridinyl)-4-((2S)-2-methyl-4-(2-propenoyl)-1-piperazinyl)pyrido[2,3-d]pyrimidin-2(1 H)-one, according to claim 2, having the following chemical structure



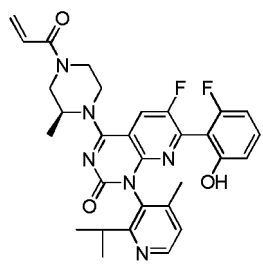
corresponding to the 1st-eluting isomer obtainable by separating the atropoisomers with supercritical fluid chromatography applying Chiralpak IC, 30x250 mm, 5 μ m, 55%MeOH/CO₂, 120g/min, 10.3 MPa (103 bar), or a pharmaceutically acceptable salt thereof.

4. Atropisomer 1 of 6-fluoro-7-(2-fluoro-6-hydroxyphenyl)-1-(4-methyl-2-(2-propenyl)-3-pyridinyl)-4-((2S)-2-methyl-4-(2-propenoyl)-1-piperazinyl)pyrido[2,3-d]pyrimidin-2(1H)-one, according to claim 2, having the following chemical structure



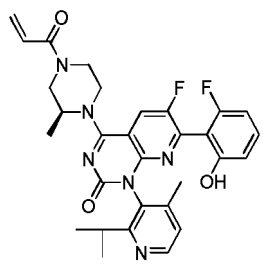
corresponding to the 2nd-eluting isomer obtainable by separating the atropoisomers with supercritical fluid chromatography applying Chiralpak IC, 30x250 mm, 5 μ m, 55%MeOH/CO₂, 120g/min, 10.3 MPa (103 bar).

5. Atropisomer 2 of 6-fluoro-7-(2-fluoro-6-hydroxyphenyl)-1-(4-methyl-2-(2-propenyl)-3-pyridinyl)-4-((2S)-2-methyl-4-(2-propenoyl)-1-piperazinyl)pyrido[2,3-d]pyrimidin-2(1H)-one, according to claim 2, having the following chemical structure



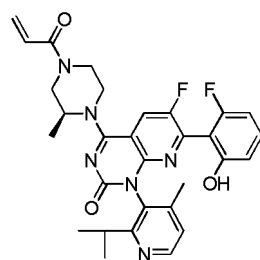
corresponding to the 2nd-eluting isomer obtainable by separating the atropoisomers with supercritical fluid chromatography applying Chiralpak IC, 30x250 mm, 5 μ m, 55%MeOH/CO₂, 120g/min, 10.3 MPa (103 bar), or a pharmaceutically acceptable salt thereof.

6. Atropisomer 2 of 6-fluoro-7-(2-fluoro-6-hydroxyphenyl)-1-(4-methyl-2-(2-propenyl)-3-pyridinyl)-4-((2S)-2-methyl-4-(2-propenoyl)-1-piperazinyl)pyrido[2,3-d]pyrimidin-2(1H)-one, according to claim 2, having the following chemical structure



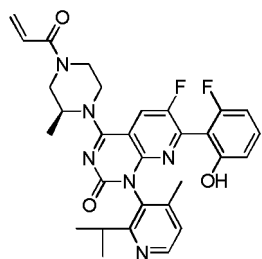
corresponding to the 2nd-eluting isomer obtainable by separating the atropoisomers with supercritical fluid chromatography applying Chiralpak IC, 30x250 mm, 5 μ m, 55%MeOH/CO₂, 120g/min, 10.3 MPa (103 bar).

7. 6-Fluoro-7-(2-fluoro-6-hydroxyphenyl)-1-(4-methyl-2-(2-propenyl)-3-pyridinyl)-4-((2S)-2-methyl-4-(2-propenoyl)-1-piperazinyl)pyrido[2,3-d]pyrimidin-2(1H)-one, according to claim 1, having the following chemical structure



or a pharmaceutically acceptable salt thereof.

8. 6-Fluoro-7-(2-fluoro-6-hydroxyphenyl)-1-(4-methyl-2-(2-propenyl)-3-pyridinyl)-4-((2S)-2-methyl-4-(2-propenoyl)-1-piperazinyl)pyrido[2,3-d]pyrimidin-2(1H)-one, according to claim 1, having the following chemical structure

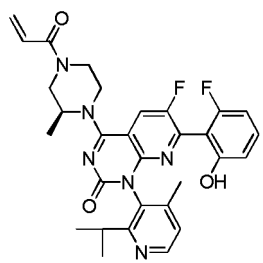


9. The compound according to any one of claims 1 to 3, 5 and 7 in the form of a pharmaceutically acceptable salt.
10. A pharmaceutical formulation comprising any one or more of the compounds of claims 1 to 9 and a pharmaceutically acceptable excipient.
11. An in vitro method of inhibiting KRAS G12C in a cell, comprising contacting the cell with the compound of any one of claims 1 to 8 or the formulation of claim 10.
12. A compound according to any one of claims 1 to 9 for use as a medicament.
13. A compound for use according to claim 12 in the treatment of cancer.
14. The compound for use according to claim 13, wherein the cancer is a hematologic malignancy, lung cancer, pancreatic cancer, endometrial cancer, appendix cancer, skin cancer, intraocular melanoma, or colorectal cancer.
15. The compound for use according to claim 14, wherein the cancer is a non-small cell lung cancer.

16. The compound for use according to any one of claims 13 to 15, wherein the cancer is a KRAS G12C mutated cancer.

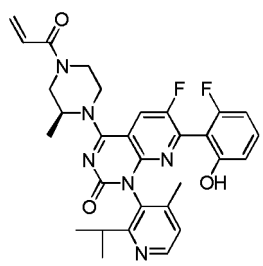
Patentansprüche

1. 6-Fluor-7-(2-fluor-6-hydroxyphenyl)-1-(4-methyl-2-(2-propenyl)-3-pyridinyl)-4-((2S)-2-methyl-4-(2-propenoyl)-1-piperazinyl)pyrido[2,3-d]pyrimidin-2(1H)-on mit der folgenden chemischen Struktur:



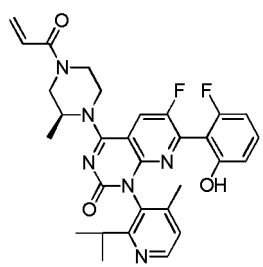
oder ein Atropisomer davon, ein pharmazeutisch akzeptables Salz davon oder ein pharmazeutisch akzeptables Salz des Atropisomers davon.

2. 6-Fluor-7-(2-fluor-6-hydroxyphenyl)-1-(4-methyl-2-(2-propenyl)-3-pyridinyl)-4-((2S)-2-methyl-4-(2-propenoyl)-1-piperazinyl)pyrido[2,3-d]pyrimidin-2(1H)-on gemäß Anspruch 1 mit der folgenden chemischen Struktur:



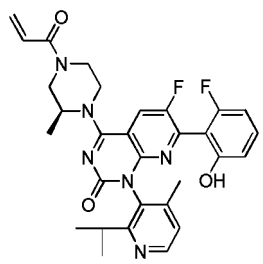
oder ein Atropisomer davon oder ein pharmazeutisch akzeptables Salz des Atropisomers davon.

3. Atropisomer 1 von 6-Fluor-7-(2-fluor-6-hydroxyphenyl)-1-(4-methyl-2-(2-propenyl)-3-pyridinyl)-4-((2S)-2-methyl-4-(2-propenoyl)-1-piperazinyl)pyrido[2,3-d]pyrimidin-2(1H)-on gemäß Anspruch 2 mit der folgenden chemischen Struktur:



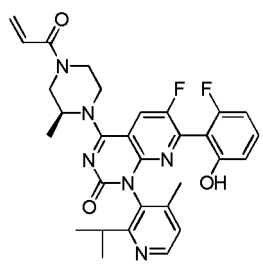
entsprechend dem ersten eluierenden Isomer, das durch das Trennen der Atropisomere mit überkritischer Flüssigkeitschromatographie unter Anwendung von Chiralpak IC, 30 x 250 mm, 5 µm, 55 % MeOH/CO₂, 120 g/min, 10,3 MPa (103 bar) erhalten werden kann, oder ein pharmazeutisch akzeptables Salz davon.

4. Atropisomer 1 von 6-Fluor-7-(2-fluor-6-hydroxyphenyl)-1-(4-methyl-2-(2-propenyl)-3-pyridinyl)-4-((2S)-2-methyl-4-(2-propenoyl)-1-piperazinyl)pyrido[2,3-d]pyrimidin-2(1H)-on gemäß Anspruch 2 mit der folgenden chemischen Struktur:



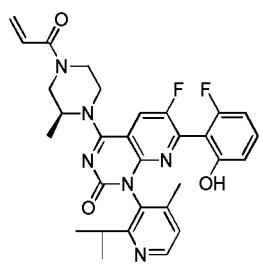
entsprechend dem ersten eluierenden Isomer, das durch das Trennen der Atropisomere mit überkritischer Flüssigkeitschromatographie unter Anwendung von Chiralpak IC, 30 x 250 mm, 5 μ m, 55 % MeOH/CO₂, 120 g/min, 10,3 MPa (103 bar) erhalten werden kann.

5. Atropisomer 2 von 6-Fluor-7-(2-fluor-6-hydroxyphenyl)-1-(4-methyl-2-(2-propenyl)-3-pyridinyl)-4-((2S)-2-methyl-4-(2-propenoyl)-1-piperazinyl)pyrido[2,3-d]pyrimidin-2(1H)-on gemäß Anspruch 2 mit der folgenden chemischen Struktur:



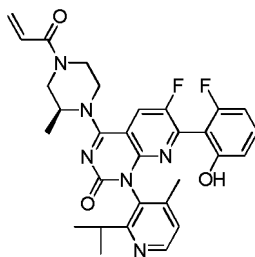
entsprechend dem zweiten eluierenden Isomer, das durch das Trennen der Atropisomere mit überkritischer Flüssigkeitschromatographie unter Anwendung von Chiralpak IC, 30 x 250 mm, 5 μ m, 55 % MeOH/CO₂, 120 g/min, 10,3 MPa (103 bar) erhalten werden kann, oder ein pharmazeutisch akzeptables Salz davon.

6. Atropisomer 2 von 6-Fluor-7-(2-fluor-6-hydroxyphenyl)-1-(4-methyl-2-(2-propenyl)-3-pyridinyl)-4-((2S)-2-methyl-4-(2-propenoyl)-1-piperazinyl)pyrido[2,3-d]pyrimidin-2(1H)-on gemäß Anspruch 2 mit der folgenden chemischen Struktur:



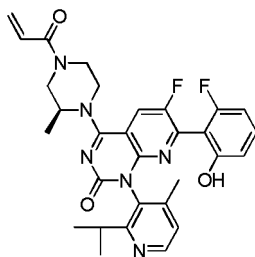
entsprechend dem zweiten eluierenden Isomer, das durch das Trennen der Atropisomere mit überkritischer Flüssigkeitschromatographie unter Anwendung von Chiralpak IC, 30 x 250 mm, 5 μ m, 55 % MeOH/CO₂, 120 g/min, 10,3 MPa (103 bar) erhalten werden kann.

7. 6-Fluor-7-(2-fluor-6-hydroxyphenyl)-1-(4-methyl-2-(2-propenyl)-3-pyridinyl)-4-((2S)-2-methyl-4-(2-propenoyl)-1-piperazinyl)pyrido[2,3-d]pyrimidin-2(1H)-on gemäß Anspruch 1 mit der folgenden chemischen Struktur:



oder ein pharmazeutisch akzeptables Salz davon.

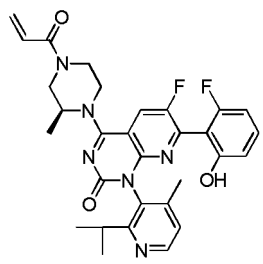
8. 6-Fluor-7-(2-fluor-6-hydroxyphenyl)-1-(4-methyl-2-(2-propenyl)-3-pyridinyl)-4-((2S)-2-methyl-4-(2-propenoyl)-1-piperazinyl)pyrido[2,3-d]pyrimidin-2(1H)-on gemäß Anspruch 1 mit der folgenden chemischen Struktur:



9. Verbindung gemäß einem der Ansprüche 1 bis 3, 5 und 7 in der Form eines pharmazeutisch akzeptablen Salzes.
10. Eine pharmazeutische Formulierung, beinhaltend eine oder mehrere der Verbindungen gemäß den Ansprüchen 1 bis 9 und einen pharmazeutisch akzeptablen Hilfsstoff.
11. Ein In-vitro-Verfahren des Inhibierens von KRAS G12C in einer Zelle, beinhaltend das In-Kontakt-Bringen der Zelle mit der Verbindung gemäß einem der Ansprüche 1 bis 8 oder der Formulierung gemäß Anspruch 10.
12. Verbindung gemäß einem der Ansprüche 1 bis 9 zur Verwendung als ein Arzneimittel.
13. Verbindung zur Verwendung gemäß Anspruch 12 bei der Behandlung von Krebs.
14. Verbindung zur Verwendung gemäß Anspruch 13, wobei der Krebs eine hämatologische Malignität, Lungenkrebs, Bauchspeicheldrüsenkrebs, Endometriumkrebs, Appendixkrebs, Hautkrebs, ein Augenmelanom oder Kolorektalkrebs ist.
15. Verbindung zur Verwendung gemäß Anspruch 14, wobei der Krebs nicht-kleinzelliger Lungenkrebs ist.
16. Verbindung zur Verwendung gemäß einem der Ansprüche 13 bis 15, wobei der Krebs KRAS-G12C-mutierter Krebs ist.

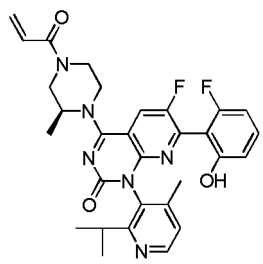
Revendications

1. 6-Fluoro-7-(2-fluoro-6-hydroxyphényl)-1-(4-méthyl-2-(2-propenyl)-3-pyridinyl)-4-((2S)-2-méthyl-4-(2-propénoyl)-1-pipérazinyl)pyrido[2,3-d]pyrimidin-2(1H)-one ayant la structure chimique suivante



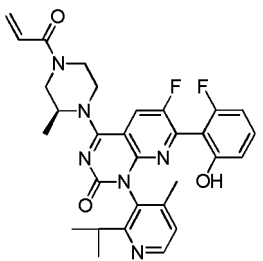
ou un atropisomère de celui-ci, un sel pharmaceutiquement acceptable de celui-ci, ou un sel pharmaceutiquement acceptable de l'atropisomère de celui-ci.

2. 6-Fluoro-7-(2-fluoro-6-hydroxyphényl)-1-(4-méthyl-2-(2-propenyl)-3-pyridinyl)-4-((2S)-2-méthyl-4-(2-propénoyl)-1-pipérazinyl)pyrido[2,3-d]pyrimidin-2(1H)-one, selon la revendication 1, ayant la structure chimique suivante



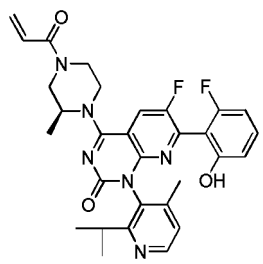
ou un atropisomère de celui-ci, ou un sel pharmaceutiquement acceptable de l'atropisomère de celui-ci.

3. Atropisomère 1 du 6-fluoro-7-(2-fluoro-6-hydroxyphényl)-1-(4-méthyl-2-(2-propenyl)-3-pyridinyl)-4-((2S)-2-méthyl-4-(2-propénoyl)-1-pipérazinyl)pyrido[2,3-d]pyrimidin-2(1H)-one, selon la revendication 2, ayant la structure chimique suivante



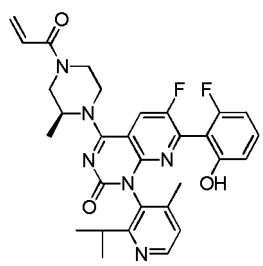
correspondant à l'isomère éluant en premier pouvant être obtenu par séparation des atropisomères à l'aide d'une chromatographie en fluide supercritique en appliquant une Chiralpak IC, 30 x 250 mm, 5 µm, 55%MeOH/CO₂, 120 g/min, 10,3 MPa (103 bars), ou un sel pharmaceutiquement acceptable de celui-ci.

4. Atropisomère 1 du 6-fluoro-7-(2-fluoro-6-hydroxyphényl)-1-(4-méthyl-2-(2-propenyl)-3-pyridinyl)-4-((2S)-2-méthyl-4-(2-propénoyl)-1-pipérazinyl)pyrido[2,3-d]pyrimidin-2(1H)-one, selon la revendication 2, ayant la structure chimique suivante



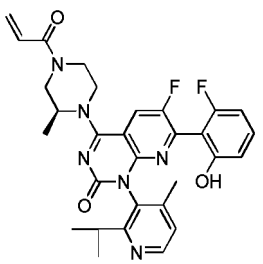
correspondant à l'isomère éluant en premier pouvant être obtenu par séparation des atropisomères à l'aide d'une chromatographie en fluide supercritique en appliquant une Chiralpak IC, 30 x 250 mm, 5 μ m, 55%MeOH/CO₂, 120 g/min, 10,3 MPa (103 bars).

5. Atropisomère 2 du 6-fluoro-7-(2-fluoro-6-hydroxyphényl)-1-(4-méthyl-2-(2-propanyl)-3-pyridinyl)-4-((2S)-2-méthyl-4-(2-propénoyl)-1-pipérazinyl)pyrido[2,3-d]pyrimidin-2(1H)-one, selon la revendication 2, ayant la structure chimique suivante



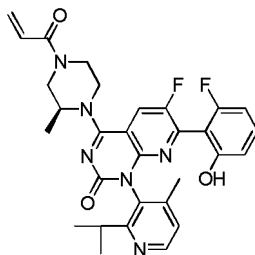
correspondant à l'isomère éluant en second pouvant être obtenu par séparation des atropisomères à l'aide d'une chromatographie en fluide supercritique en appliquant une Chiralpak IC, 30 x 250 mm, 5 μ m, 55%MeOH/CO₂, 120 g/min, 10,3 MPa (103 bars), ou un sel pharmaceutiquement acceptable de celui-ci.

6. Atropisomère 2 du 6-fluoro-7-(2-fluoro-6-hydroxyphényl)-1-(4-méthyl-2-(2-propanyl)-3-pyridinyl)-4-((2S)-2-méthyl-4-(2-propénoyl)-1-pipérazinyl)pyrido[2,3-d]pyrimidin-2(1H)-one, selon la revendication 2, ayant la structure chimique suivante



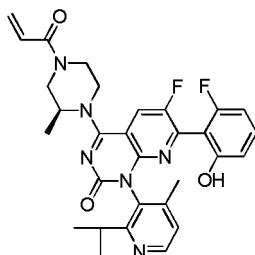
correspondant à l'isomère éluant en second pouvant être obtenu par séparation des atropisomères à l'aide d'une chromatographie en fluide supercritique en appliquant une Chiralpak IC, 30 x 250 mm, 5 μ m, 55%MeOH/CO₂, 120 g/min, 10,3 MPa (103 bars).

7. 6-Fluoro-7-(2-fluoro-6-hydroxyphényl)-1-(4-méthyl-2-(2-propanyl)-3-pyridinyl)-4-((2S)-2-méthyl-4-(2-propénoyl)-1-pipérazinyl)pyrido[2,3-d]pyrimidin-2(1H)-one, selon la revendication 1, ayant la structure chimique suivante



ou un sel pharmaceutiquement acceptable de celui-ci.

8. 6-Fluoro-7-(2-fluoro-6-hydroxyphényl)-1-(4-méthyl-2-(2-propényl)-3-pyridinyl)-4-((2S)-2-méthyl-4-(2-propényl)-1-pipérazinyl)pyrido[2,3-d]pyrimidin-2(1H)-one, selon la revendication 1, ayant la structure chimique suivante



9. Le composé selon l'une quelconque des revendications 1 à 3, 5 et 7 se présentant sous la forme d'un sel pharmaceutiquement acceptable.
10. Une formulation pharmaceutique comprenant un quelconque ou plusieurs des composés des revendications 1 à 9 et un excipient pharmaceutiquement acceptable.
11. Un procédé in vitro d'inhibition de KRAS G12C dans une cellule, comprenant la mise en contact de la cellule avec le composé de l'une quelconque des revendications 1 à 8 ou la formulation de la revendication 10.
12. Un composé selon l'une quelconque des revendications 1 à 9 pour une utilisation en tant que médicament.
13. Un composé pour une utilisation selon la revendication 12 dans le traitement d'un cancer.
14. Le composé pour une utilisation selon la revendication 13, dans lequel le cancer est une hémopathie maligne, un cancer du poumon, un cancer du pancréas, un cancer de l'endomètre, un cancer de l'appendice, un cancer de la peau, un mélanome intra-oculaire, ou un cancer colorectal.
15. Le composé pour une utilisation selon la revendication 14, dans lequel le cancer est un cancer du poumon non à petites cellules.
16. Le composé pour une utilisation selon l'une quelconque des revendications 13 à 15, dans lequel le cancer est un cancer présentant une mutation KRAS G12C.

REFERENCES CITED IN THE DESCRIPTION

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- WO 2019213516 A1 [0004]

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